Final Project Checkpoint 1

CMSC 320 - Introduction to Data Science
Fall 2026
Instructor: Dr. Fardina Alam

Summary

For your final semester long group project, you will publish a tutorial that will walk users through the entire data science pipeline: acquiring and curating data, parsing the data into a queryable format, exploratory data analysis, hypothesis testing and machine learning, and explaining the results with words and visualizations. This project is designed to demonstrate your data science knowledge and contribute to your portfolio.

Deadlines

- **Project posted:** September 8, 2026
- **First deliverable:** September 29, 2026
- **Second deliverable:** October 29, 2026
- **Final deliverable:** December 10, 2026

Instructions for later deliverables will be posted no later than 3 weeks before the date they are due. Late submissions will **not** be accepted for the final deliverable.

Checkpoint 1 Rubric (5 points)

Submit a **pdf** on Gradescope with:

- **(1 point)** A **link** to your Github repository.
- **(2 points)** **What** datasets are you choosing? Cite the source(s).
    - The dataset should be large enough and appropriate for making your analysis.
- **(2 points)** **Why** are you choosing this dataset?

Dataset Ideas

Choose an application area or dataset that is of interest to you. Please feel free to be creative! Remember that you can use API calls or scrape websites for data.

There are lots of options for large, open-access datasets that could yield multiple insights, especially from governmental agencies. Here are some examples:

- Canada has published some pretty interesting statistics – check the left hand side for filtering options: https://www.statcan.gc.ca/en/start. One of their datasets is the retail

price of products over time: https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=1810024501
- FEC campaign finance data: https://www.fec.gov/data/
- USGS earthquake data: https://earthquake.usgs.gov/earthquakes/search/

You could also curate a personalized dataset. Keep in mind that the dataset needs to be large and varied enough for machine learning analysis.

- Is your data being tracked by a website? For example, YouTube watch history or Goodreads book histories. Check whether you can export it. (Only do this if you're willing to discuss the trends within – not everyone is comfortable with that).
- What are your interests? What do you like to watch? Is there a subreddit for it that hosts a survey every year, or a fan-made spreadsheet?
- Generate your own dataset from audio or video data, e.g. https://www.reddit.com/r/educationalgifs/comments/64rjns/i_did_a_center_of_mass_analysis_of_a_triple/
    - Relatedly, what have you learned about in your other classes that you would want to demonstrate through data analysis?